Puram 2017 HNSCC — recapitulation summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

10/17 samples (≥100 cells) intermix on one continuous manifold; no per-patient islands (64 detached cells removed).

Gate 2 — module geography → GO

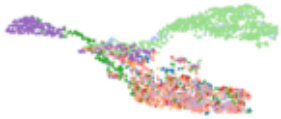
Gast/pSom hub diverges into arms; Crest is a terminal readout at the Basal tip (0.43 > Atypical 0.27 ≈ Classical 0.24).

Gate 3 — hub mutual exclusivity → GO

On 466 Mixed-hub cells, per-gene Gast vs pSom correlation vectors anti-correlate: r = −0.47 (ref ≈ −0.49).

STEP FIGURES

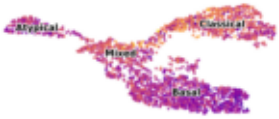
Gate 1: integration



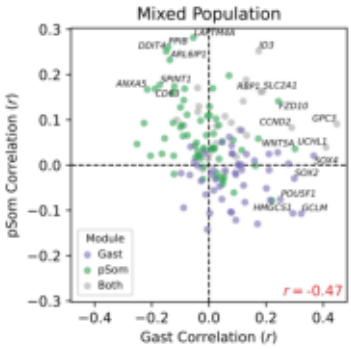
Trajectory / root



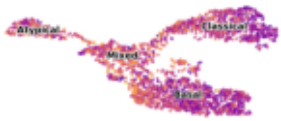
Gast



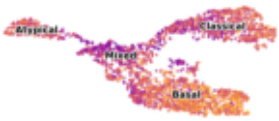
Gate 3: r = −0.47



pSom



Crest



EMT-TFs



Dataset.

Head & Neck SCC (Puram et al., 2017), SmartSeq2 — 18 patients / 5,902 cells (master scope); malignant only: 2,539 load, 2,408 across 4 subtypes enter the trajectory.

Verdict.

✓ Pattern reproduces — all three gates pass. Gast/pSom mutually exclusive within the Mixed hub (r = −0.47 vs ref −0.49), hub at the root of a hub-and-spoke manifold; EMT-TFs ZEB1 (Classical) / PRRX1 (Atypical).

Parameter choices

Min cells/sample	≥ 100 (keeps 10/17)	**Integration**	Harmony on sample, 20 it
Island removal	UMAP radius-graph (−64 cells)	**Trajectory**	scFates PPT N=200 λ=2000 σ=0.5
Root node	85 = Mixed hub (466 cells)	**Subtypes**	markers + Puram 2017
EMT-TFs shown	ZEB1 (Classical) / PRRX1 (Atypical)	**Reference r**	−0.49 (embryonic anchor)